

PSP Database Optimisation Analysis

Scope

This document focuses on identifying opportunities to improve database performance and take actions accordingly by troubleshooting the root cause.

Introduction

There are multiple multi-year business goals like Grow Workforce and Equifax, which have quarterly milestones to achieve. This will bring additional data and workload to the PSP monolith database. To support these business goals, we need to ensure the PSP monolith database is performing well and has enough headroom of resources to accommodate future increases in workload.

Problem statement

There is high usage in ReadIOPS and ReadThroughput observed in the monolith database. GREEN

Analysis

Analysis - 24-Sep-21 19:00 - 24-Sep-21 19:45

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
31,561,112	1	31,561,112.00	83.58	1,147.38	67.17	33.93	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_STA...	Query Details GemsAccountsReceivableProcessor

Analysis - 24-Sep-21 18:30 - 24-Sep-21 19:00

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
53,553,913	0		63.41	381.61	46.92	54.18	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_STA...	Query Details GemsAccountsReceivableProcessor
4,718,576	1	4,718,576.00	5.59	100.30	17.11	25.04	fud06g9jzyh2	BaseATFExtractFileProcess:146	/* Select companyDailyLiabilit...	Query BaseATFExtractFileProcess (companyDailyLiabilit)
4,140,680	1	4,140,680.00	4.90	376.26	13.15	79.19	5nmd8a96rcf9j	BaseATFExtractFileProcess:146	/* Select ptp.Law , ptp.Amount...	Query BaseATFExtractFileProcess Select ptp.Law
2,252,812	1	2,252,812.00	2.67	231.97	8.20	93.73	5ttnr2d161kga	BaseATFExtractFileProcess:146	/* select co, serviceInfo.Stat...	Query BaseATFExtractFileProcess(select co, serviceInfo.Stat)
1,286,568	1	1,286,568.00	1.52	159.00	5.51	87.57	a1zfnguds0wgx	BaseATFExtractFileProcess:146	/* select comp. SourceCompanyld...	Query BaseATFExtractFileProcess (select comp. SourceCompanyld)
904,345	1	904,345.00	1.07	538.02	16.38	91.20	2nd0g8cwbz6zg	DailyBatchJobsProcessor:776	BEGIN prc_offload_update_payro...	Based On Query BaseATFExtractFileProcess (select comp. SourceCompanyld)
878,862	1	878,862.00	1.04	509.02	14.20	93.51	1w9qy5w5y3cxf	DailyBatchJobsProcessor:776	UPDATE PSP_PAYROLL_RUN PR SET ...	Query DailyBatchJobsProcessor (UPDATE PSP_PAYROLL_RUN PR SET ...)

Analysis - 23-Sep-21 19:00 - 23-Sep-21 19:45

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
85,082,230	1	85,082,230.00	84.92	1,505.50	62.41	38.50	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_STA...	Query Details GemsAccountsReceivableProcessor
2,793,936	1	2,793,936.00	2.79	131.11	41.74	67.78	5vfmc964z0tqr	ViewMyPaycheckAdapter:232	/* criteria query */ select co...	Query ViewMyPaycheckAdapter(/* criteria query */ select co...)
1,289,960	1	1,289,960.00	1.29	332.87	27.91	76.05	2su4my5wrmmbv	ViewMyPaycheckAdapter:199	select * from (/* criteria qu...	Query ViewMyPaycheckAdapter(select * from (/* criteria qu...)

Analysis - 23-Sep-21 18:30 - 23-Sep-21 19:00

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
4,715,710	1	4,715,710.00	12.35	99.83	19.88	42.17	39f2687k24czy	BaseATFExtractFileProcessor:146	/* Select companyDailyLiabilit...	Query BaseATFExtractFileProcess (companyDailyLiabilit)
1,503,359	1	1,503,359.00	3.94	860.18	15.24	91.30	2nd0g8cw bz6zg	DailyBatchJobsProcessor:776	BEGIN prc_offload_update_payro...	Based On Query BaseATFExtractFileProcess (select comp. SourceCompanyId)
1,429,802	1	1,429,802.00	3.74	792.92	12.71	94.02	1w9qy5w5 y3cxf	DailyBatchJobsProcessor:776	UPDATE PSP_PAYROLL_RUN PR SET ...	Query DailyBatchJobsProcessor (UPDATE PSP_PAYROLL_RUN PR SET ...)
591,873	1	591,873.00	1.55	338.75	14.63	89.22	9zdchram sws6b	DailyBatchJobsProcessor:776	INSERT /*+ APPEND */ INTO PSP_...	Query DailyBatchJobsProcessor(INSERT /*+ APPEND */ INTO PSP_...)
591,873	1	591,873.00	1.55	338.75	14.63	89.22	fx6bn8avj 7cp2	DailyBatchJobsProcessor:776	BEGIN PRC_OFFLOAD_INSERT_F TS(...	Based On Query DailyBatchJobsProcessor(INSERT /*+ APPEND */ INTO PSP_...)
498,713	1	498,713.00	1.31	86.50	5.10	95.87	25kf4y2hs gf1x	BaseATFExtractFileProcessor:146	/* select c. SourceCompanyId, c...	
405,821	1	405,821.00	1.06	75.59	19.96	89.15	bbd4wbqn zr8p6	ATFDataExtractProcessor:497	/* named HQL query findCompani...	

Analysis - 22-Sep-21 19:00 - 22-Sep-21 19:45

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
44,813,170	0		52.76	319.75	46.19	54.83	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_S TA...	Query Details GemsAccountsReceivableProcessor
4,713,495	1	4,713,495.00	5.55	99.97	17.71	38.75	fud06g9jzyh2	BaseATFExtractFileProcessor:146	/* Select companyDailyLiabilit...	Query BaseATFExtractFileProcess (companyDailyLiabilit)
3,289,382	1	3,289,382.00	3.87	156.61	9.89	68.46	5nmd8a96rcf9j	BaseATFExtractFileProcessor:146	/* Select ptp.Law , ptp. Amount...	Query BaseATFExtractFileProcess Select ptp.Law
2,324,485	1	2,324,485.00	2.74	282.61	7.75	94.36	5ttnr2d161kga	BaseATFExtractFileProcessor:146	/* select co, serviceInfo.Stat...	Query BaseATFExtractFileProcess(select co, serviceInfo.Stat)
1,903,416	1	1,903,416.00	2.24	1,236.51	16.76	86.97	9zdchram sws6b	DailyBatchJobsProcessor:776	INSERT /*+ APPEND */ INTO PSP_...	Query DailyBatchJobsProcessor(INSERT /*+ APPEND */ INTO PSP_...)
1,903,416	1	1,903,416.00	2.24	1,236.51	16.76	86.97	fx6bn8avj 7cp2	DailyBatchJobsProcessor:776	BEGIN PRC_OFFLOAD_INSERT_F TS(...	Based On Query DailyBatchJobsProcessor(INSERT /*+ APPEND */ INTO PSP_...)
1,300,244	1	1,300,244.00	1.53	161.06	5.44	88.49	a1zfnguds 0wgx	BaseATFExtractFileProcessor:146	/* select comp. SourceCompanyId...	Query BaseATFExtractFileProcess (select comp. SourceCompanyId)

Analysis - 21-Sep-21 19:00 - 21-Sep-21 19:45

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
11,143,921	1	11,143,921.00	52.68	956.35	72.54	28.14	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_STA...	Query Details GemsAccountsReceivableProcessor

Analysis - 21-Sep-21 18:30 - 21-Sep-21 19:00

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
6,617,455	1	6,617,455.00	5.95	1,110.69	15.59	89.10	5nmd8a96rcf9j	BaseATFExtractFileProcessor:146	/* Select ptp.Law , ptp. Amount...	Query BaseATFExtractFileProcess Select ptp.Law
4,713,495	1	4,713,495.00	4.24	106.27	14.73	84.23	fud06g9jzyh2	BaseATFExtractFileProcessor:146	/* Select companyDailyLiabilit...	Query BaseATFExtractFileProcess (companyDailyLiabilit)
2,501,012	1	2,501,012.00	2.25	200.03	7.91	93.61	5ttnr2d161kga	BaseATFExtractFileProcessor:146	/* select co, serviceInfo. Stat...	Query BaseATFExtractFileProcess(select co, serviceInfo.Stat)
1,269,865	1	1,269,865.00	1.14	153.67	6.96	94.24	a1zfnguds 0wgx	BaseATFExtractFileProcessor:146	/* select comp. SourceCompanyId...	Query BaseATFExtractFileProcess (select comp. SourceCompanyId)

Analysis - 20-Sep-21 19:00 - 20-Sep-21 19:45

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
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8,791,805	1	8,791,805.00	52.09	898.46	71.21	29.66	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_STA...	Query Details GemsAccountsReceivableProcessor
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Analysis - 20-Sep-21 18:30 - 20-Sep-21 19:00

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	% IO	SQL Id	SQL Module	SQL Text	SQL Details
76,127,056	0		70.15	592.00	50.35	50.60	6qk524uazdf0c	GemsAccountsReceivableProcessor:	UPDATE PSP_FINANCIAL_TRANS_STA...	Query Details GemsAccountsReceivableProcessor
4,713,495	1	4,713,495.00	4.34	99.85	17.02	25.37	fud06g9jzyyh2	BaseATFExtractFileProcess:146	/* Select companyDailyLiabilit...	Query BaseATFExtractFileProcess (companyDailyLiabilit)
3,415,538	1	3,415,538.00	3.15	192.68	10.64	70.72	5nmd8a96rcf9j	BaseATFExtractFileProcess:146	/* Select ptp.Law , ptp.Amount...	Query BaseATFExtractFileProcess Select ptp.Law
2,515,997	1	2,515,997.00	2.32	212.07	7.74	93.06	5ttnr2d161kga	BaseATFExtractFileProcess:146	/* select co, serviceInfo.Stat...	Query BaseATFExtractFileProcess(select co, serviceInfo.Stat)
1,271,218	1	1,271,218.00	1.17	160.79	6.87	87.16	a1zfnguds0wgx	BaseATFExtractFileProcess:146	/* select comp. SourceCompanyId...	Query BaseATFExtractFileProcess (select comp.SourceCompanyId)

Queries To Optimise

- [Query BaseATFExtractFileProcess\(/* dynamic native SQL...](#))
- [Query BaseATFExtractFileProcess\(companyDailyLiabilit\)](#)
- [Query BaseATFExtractFileProcess\(select co, serviceInfo.Stat\)](#)
- [Query BaseATFExtractFileProcess \(select comp.SourceCompanyId\)](#)
- [Query BaseATFExtractFileProcess Select ptp.Law](#)
- [Query DailyBatchJobsProcessor\(INSERT /*+ APPEND */ INTO PSP_...](#))
- [Query DailyBatchJobsProcessor \(UPDATE PSP_PAYROLL_RUN PR SET ...\)](#)
- [Query Details GemsAccountsReceivableProcessor](#)
- [Query ViewMyPaycheckAdapter\(/* criteria query */ select co...\)](#)
- [Query ViewMyPaycheckAdapter\(select * from \(/* criteria qu...\)](#)
- [Query WorkersCompGatewayManager\(/* criteria query */ select\)](#)